Q488 converse at first order in q — proof file (verbatim)

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Title	Q488 converse at first order in q: what's true, what's false, what's proved
WIP commit	`grandpa-rick/work-in-progress @ WIPHASH`
Source	`proofs/2026-10-18-q488-converse-order-q1.md` (copied verbatim, with three combining accents precomposed for pdflatex, as `reviews/2026-10-17-q488-converse-order-q1-proofs.md`)
Registry	`registry/q488-converse.json`

This is the Markdown proof file set verbatim, with no edits. The companion note `2026-10-17-q488-converse-order-q1.pdf` is the typeset summary. One correction to the file's addendum: the $N = 5$ $q = \infty$ run it calls "5/12, rest running" has since finished, with 12/12 first order at $q = \infty$ on all 26 predicted pairs (`scripts/day243/q488c/qinf_N5_partial.txt`).

Q488 converse at first order in q -- what's true, what's false, what's proved

Rick, Day 243 PROVE (session dated 2026-10-17; file name per PROVE.md).
Scripts: `work-in-progress/scripts/day243/q488c/`.
Source: Motegi, arXiv:2610.15344, e-print source `typebcweightfunctions_arxivversionfinal-3.tex`. I deep-read L474-487 (Thm 3.3, the symmetrisation formula), L1057-1062 (E^C), L603-620 (Thm 2.6(i), triangularity and the diagonal) and L2641-2848 (stability) first-hand today. Clio's Q488 tex (0e59afb) supplies notation.

0. Verdict in one screen

PROVE asked for (a)-(d). **Two of the three main targets are FALSE as stated, and both fail first at N = 4.**

item	statement	status
(a)	$c \in C(w)[[q]]$	**PROVED**, all N (§2)
(b)	jump $\geq 2 \Rightarrow c_0 = 0$	**FALSE** at N = 4: c_0^{1234}_{12,34} = $1/(w_3^2 w_4^2)$. 17 jump-$\geq$2 triples at N=4 have $c_0 \neq 0$. True for $N \leq 3$ (exhaustive)
(c)	jump $\geq 2 \Rightarrow c_1$ has a long-root pole	**FALSE** at N = 4: $(\lambda,\mu,\nu) = (\varnothing,\{3\},\{1,2,4\})$ has c_1 Laurent on every H_ab. Its full-q residue on H_{12} is $\neq 0$ but has q-valuation **2**. Holds at $N \leq 3$ (13/13), and for 126/127 jump-$\geq$2 triples at N=4
(c-top)	if ν minus its two largest elements is $\geq \lambda$, μ, then c_1^ν has a simple pole on H_{top two}	**PROVED**, all N, all jumps (§6, Thm 9)
(c-agg)	for each (λ,μ): some $c^\nu_{\lambda\mu}$ (some ν) has a pole on H_ab $\Leftrightarrow \exists$ M'' $\subseteq$ [N]\{a,b} with M'' $\geq \lambda$, μ	**PROVED**, all N, full q (§4, Thm 6)
(d)	jump $\leq 1 \Rightarrow c_1$ Laurent	**PROVED**, all N (§6bis, Thm 10). Also c_0 is Laurent for every triple. Computed: $N \leq 3$ full; N = 4, 0 long-root poles in 599 jump-$\leq$1 triples on all six pairs

So **the order-q^1 route cannot settle Clio's converse (Gap 2) in general**. It settles the converse exactly on the "top-two" family, which includes every $(\varnothing,\varnothing,\nu)$ with $|\nu| \geq 2$, the N=2 witness, and 6/13 of the N=3 jump-$\geq$2 triples; with the single-defect theorem (§6, Thm 9') all 13 at N=3 and 111/127 at N=4. The missing triples are those whose created pole sits in a *high block* (§7). There the q^1 coefficient of the residue can vanish identically while the full-q residue survives.

Engine of everything: **Lemma QC (§3)**, a q-twisted collision on the long-root divisors, proved from Motegi's formula:
> a<b $\in$ M, on H_ab = {w_a w_b = 1}: $\tilde{A}$_{M,I} = g(q,w_a) $\cdot$ $\tilde{A}$_{M\{a,b},I} for every I,
> $g(q,x) = (1-q)^2 / ((1-qx^2)(1-qx^{-2}))$.

At q=0 we have g=1: that's GKM, and the q=0 basis is a K_T(LG(N,2N))-type Schubert basis. At first order, $g' = (x - x^{-1})^2 =: \gamma$. Clio's long-root poles are exactly the failure of the twisted GKM condition to be untwisted.

1. Setup and the facts I take from Motegi

φ_I(M) := $\tilde{A}$_{M,I} := $\widetilde{W}$C_{N,n,|I|}(w_M|w|I), for any n $\geq$ |M|, |I| (n-independent by stability, Motegi L2823-2848).

The c^ν_{$\lambda\mu$} are defined by $\varphi_\lambda\varphi_\mu = \Sigma_\nu$ c^ν_{$\lambda\mu$}φ_ν (pointwise on the 2^N points M).
M $\geq$ I is Motegi's order (|M| $\geq$ |I|, and the top |I| elements of M dominate I entrywise).
Notation: M'' := M\{a,b}; h := 1 $-$ 1/(w_a w_b) is the local parameter of H = H_ab; O_H is the local ring.

Used from Motegi (all deep-read today):
- (M1) the symmetrisation formula (Thm 3.3, L474-487). WC = F at β=1, γ=0. F is a Laurent polynomial in u, w and a
  polynomial in q (a symmetrised rational function with at most simple poles on u_j = u_k, which cancel).
- (M2) E^C (L1057-1062); $\tilde{A}$ = WC/E^C.
- (M3) Triangularity: $\tilde{A}$_{M,I} = 0 unless M $\geq$ I (Thm 2.6(i)).
- (M4) The diagonal $\tilde{A}$_{I,I} is Clio's eq:diag: a product over all positive roots of C_N. Long-root factors (1 $-$ w_{i}^{-1}w_{i'}^{-1}) for i<i' in I each occur once. (Q)-factors (1 $-$ q·monomial). No other factor vanishes
  identically on any H_ab for generic q. At q=0 the diagonal is the product of the L, S, A factors, and it is $\neq$ 0.
- (M5) Stability (L2641-2647, L2823-2848), so we may take n = |M| ("unpadded", all u-values distinct).

2. (a): c $\in$ C(w)[[q]] -- PROVED

R := C[q, w^$\pm$]_{(q)} is a DVR with residue field C(w), and R $\subset$ C(w)[[q]].
By (M1) WC $\in$ C[q][w^$\pm$] after specialisation. Every factor of E^C is (1$-$q)^{$\pm$1} or (1 $-$ q·monomial)^{$\pm$1}, a unit in R.
So $\tilde{A}$_{M,I} $\in$ R. By (M3), det($\tilde{A}$) = $\pm$ Π_I $\tilde{A}$_{I,I}. Each $\tilde{A}$_{I,I} mod q is the nonzero product of its L, S, A factors
(M4), so it is a unit of R, and hence $\tilde{A}$^{-1} has entries in R. Then c^$\bullet$_{$\lambda\mu$} = $\tilde{A}$^{-1}($\varphi_\lambda\varphi_\mu$) $\in$ R^{2^N}. ■

Consequence. c_0 := c|_{q=0} is the structure-constant tensor of the basis φ^0 := $\tilde{A}$|_{q=0}, and
c_1 = ∂_q c|_{q=0}. Write δ_I := ∂_qφ_I|_{q=0}. Differentiating $\varphi_\lambda\varphi_\mu = \Sigma$ c^ν φ_ν gives

$$\Sigma_\nu\ c_1{}^\nu\ \varphi^0{}_\nu = \Phi := \varphi^0{}_\lambda\ \delta_\mu + \delta_\lambda\ \varphi^0{}_\mu - \Sigma_\kappa\ c_0{}^\kappa\ \delta_\kappa\ . \tag{1.1}$$

Laurent at every order is necessary. If c is Laurent in w over C(q), then since c $\in$ R its coefficients are regular at q=0 (a negative q-power would leave a nonzero Laurent polynomial in w as its coefficient). So every c_k
is Laurent. Hence: **a pole of c_1 certifies that c is not Laurent.**

2bis. The explicit q = 0 formula -- PROVED

Put q=0 in (M1) with n=|M| (M5), u = w_M. Then E^C|_{q=0} = 1. Per head h (label i): (1$-$h^{-2})h^{N+1}Π_{k<i}(1 $-$ w_k/h).
Head pair: h_jh_k(1 $-$ 1/(h_jh_k))/(1 $-$ h_j/h_k). Tail t: t^{N+1}. Tail-head: th(1 $-$ 1/(th))/(1 $-$ t/h).
Tail pair: tt'. Every value collects exactly u^{N+n}, which cancels the prefactor. The (n$-$r)! tail orderings cancel the 1/(n$-$r)!. Hence, summing over injective h: [r] $\to$ M (h_j is the head carrying label κ_j):

$$\varphi^0{}_\kappa(M) = \Sigma_h\ \Pi_j\ (1-h_j{}^{-2})\ \Pi_{k<\kappa_j}(1 - w_k/h_j)\cdot \Pi_{j<k}\ (1 - 1/(h_jh_k))/(1 - h_j/h_k) \cdot \Pi_{t \in M\setminus h}\ \Pi_j\ (1 - 1/(t\ h_j))/(1 - t/h_j). \tag{2.1}$$

(Checked against the engine on all 64 (M,I) pairs at N=3: `testq0.py`.) Immediate consequences:
- **$\varphi^0{}_\varnothing \equiv 1$** (r = 0: the empty product). Hence $c_0{}^\nu{}_{\varnothing\varnothing} = \delta_{\nu\varnothing}$.
- **N-stability at q=0.** (2.1) only involves w_k for k < max κ, together with the values w_M. So for M, $\kappa \subseteq$ [N'],
  $\varphi^0{}_\kappa(M)$ does not depend on N $\geq$ N'. In particular $c_0{}^\nu{}_{\lambda\mu}$, for λ, μ, $\nu \subseteq$ [N'], does not depend on N.
- **Vanishing terms.** If h_j = w_m with m < κ_j, the factor k = m is (1 $-$ 1) = 0, identically in w (and at all q:
  the factor (u_j $-$ w_k), k < i_j, carries no q).

Lemma L (limit lemma). Let a, b $\notin$ M, and let M $\geq \kappa$. Restrict to H_ab: w_a = x, w_b = 1/x, with x fixed and
generic. Take w_c = t^{$-$e_c} for c $\notin$ {a,b}, with 0 < e_1 < e_2 < ... increasing in c, and let t $\to 0^+$. Then $\varphi^0{}_\kappa(M) \to 1$. Hence $\varphi^0{}_\kappa(M)|_H \not\equiv 0$. (Also without H: $\varphi^0{}_\kappa(M) \to$ [M $\geq \kappa$]. This is the q=0 converse of (M3).)

Proof. Every surviving term has h_j = w_m with m $\geq$ κ_j. In such a term every factor (1 $-$ X) whose X $\to$ 0 tends to 1.

That covers $(1 - h^{-2})$, $(1 - 1/(h\,h'))$, $(1 - 1/(t\,h))$, and the factors $(1 - w_k/h_j)$ with $k < \kappa_j \le m$. (For $k \in \{a,b\}$ these are $1 - x^{\pm 1}/h_j \to 1$, since $h_j \to \infty$.) The denominators $(1 - h_j/h_k)$ and $(1 - t/h_j)$ tend to 1 when the numerator variable has the smaller index, and to ∞ otherwise. So a term tends to 1 if its heads increase and every tail lies below every head, and to 0 otherwise. Exactly one such h exists: the top r elements of M, in increasing order. It is a surviving term iff $m_{s-r+j} \ge \kappa_j$ for all j, i.e. iff $M \ge \kappa$. ■

3. Lemma QC (q-twisted long-root collision), all q -- PROVED

Lemma QC. Let a < b with a, b ∈ M. On H_ab, for every I: $\tilde{A}_{M,I} = g(q, w_a) \cdot \tilde{A}_{M'',I}$, where $g = (1-q)^2/((1-qw_a^2)(1-qw_a^{-2}))$.

Proof. Use (M1) unpadded: $n = s = |M|$ for $\tilde{A}_{M,I}$, and $n = s-2$ for $\tilde{A}_{M'',I}$ (M5). Write $x := w_a$, so $w_b = 1/x$.

1. *Special values in the head die.* If both x and 1/x are heads, the head-pair factor $(u_j u_k - 1) = 0$. If one is a head and the other a tail, the tail-head factor $(u_j u_k - 1) = 0$. So only permutations that put both special values in the tail survive. Tail positions are symmetric, so after cancelling $(n-r)!$ the sum runs over ordered head selections h from the remaining $s-2$ values. These are exactly the head selections of the $(s-2)$-formula.
2. *Compare term by term* (same h, remaining tails T). The s-term is the $(s-2)$-term times the extra factors:
   - the prefactor $(x \cdot x^{-1})^{-(N+s)} \cdot \prod_{v \in \mathrm{rest}} v^{-2}$ (the change $N+s-2 \to N+s$);
   - $(\beta x - \gamma)(\beta/x - \gamma) = 1$;
   - $\prod_k (x - q\,w_k)(1/x - q\,w_k)$;
   - for each head h: $(qx - h)(xh - 1)/(x - h) \cdot (q/x - h)(h/x - 1)/(1/x - h) = (qx - h)(q/x - h) = h^2(1 - qx/h)(1 - q/(xh))$;
   - for each other tail t: $(xt - q)(t/x - q) = t^2(1 - q/(xt))(1 - qx/t)$;
   - the pair itself: $(x \cdot x^{-1} - q) = (1 - q)$.

   The x-dependence is the same for heads and tails. So the bracket equals $(1-q)\ \prod_k (x-qw_k)(1/x-qw_k)\ \prod_{v \in \mathrm{rest}} v^2(1 - qx/v)(1 - q/(xv))$, independent of h, and the v^2 cancels the prefactor. Hence $F_s(\mathrm{rest}, x, 1/x) = (1-q)\ \prod_k (x-qw_k)(x^{-1}-qw_k)\ \prod_v (1-qx/v)(1-q/(xv)) \cdot F_{s-2}(\mathrm{rest})$.

   (If $s-2 < r$, no head selection survives and both sides are 0.)
3. *Normalisation.* By (M2), $E_s/E_{s-2} = (1-qx^2)(1-qx^{-2})\ \prod_v (1-qx/v)(1-q/(xv))\ \prod_k (1-qxw_k)(1-qw_k/x)\ /\ (1-q)$.

   (The $(1-q)^{-2}$ offset against the diagonal pair factors $(1-q)^2$, and the $\prod_v (1 - q\,x\,v)(1 - q\,v/x)$ cancel between numerator and denominator.) Since $(x - qw)(x^{-1} - qw) = (1 - qxw)(1 - qw/x)$, the ratio is $(1-q)^2/((1-qx^2)(1-qx^{-2}))$. ■

Computed on all (M, pair, I) at N=2, 3 (`ratio.py`).

Corollary 4. (i) At q=0: $\varphi^0_I(M) = \varphi^0_I(M'')$ on H_ab (long-root GKM). (ii) At first order, since $g(0) = 1$ and $\partial_q\, g(0) = x^2 + x^{-2} - 2 = \gamma := (w_a - w_a^{-1})^2$: $\delta_I(M) - \delta_I(M'') = \gamma\ \varphi^0_I(M'')$ on H_ab.

4. Lattice lemma and the aggregate converse, full q -- PROVED

Fix a<b. For $g \in k(H)^\times$ let $L_g := \{f \in O_H^{2^N} : f(M) - g\,f(M'') \in hO_H \text{ for all } M \supseteq \{a,b\}\}$.

Prop 5. $\mathrm{span}_{O_H}\{\varphi_I\} = L_g$ (generic q, with g from QC), and $\mathrm{span}_{O_H}\{\varphi^0_I\} = L_1$. Hence, for any $f \in O_H^{2^N}$: every coordinate of f in the basis φ is regular on H iff $f \in L_g$. Poles are at most simple, and the polar part $\mathrm{Res}_H\, \ell_\nu(f)$ depends only on the defects $(f(M) - g\,f(M''))|_H$, linearly.

Proof. (i) ⊆: the entries lie in O_H (they are Laurent in w over C(q), or over C at q=0), and they satisfy the congruence by QC. (ii) The colength of L_g is $\#\{M \supseteq ab\} = 2^{N-2}$, since the defect map onto $k(H)^{2^{N-2}}$ is

surjective (take f = c·e_M) and has kernel L_g. (iii) The colength of the span is v_H(det $\tilde{A}$) = Σ_I
↪ v_H($\tilde{A}$_{I,I}),
and that is 2^{N−2} by (M4): exactly the I ⊇ {a,b}, each with multiplicity one. Equal colengths and an
↪ inclusion
give equality. The rest follows because h·O^{2^N} ⊆ L_g. ■

Theorem 6 (aggregate converse, all N, generic q). For fixed λ, μ and a<b, the following are equivalent:
(i) every c^ν_{$\lambda\mu$} (ν ranging over all subsets) is regular on H_ab;
(ii) φ_$\lambda\varphi$_μ(M'')|_H = 0 for every M'' ⊆ [N]\{a,b};
(iii) no M'' ⊆ [N]\{a,b} satisfies both M'' ≥ λ and M'' ≥ μ.

Proof. f := φ_$\lambda\varphi$_μ satisfies f(M) = g^2 f(M'') on H by QC. So f ∈ L_g iff (g^2 − g) f(M'')|_H = 0. Since g ≠
↪ 0, 1
on H, Prop 5 gives (i)⇔(ii). (iii)⇒(ii) is (M3). (ii)⇒(iii): if M'' ≥ λ, μ, then by Lemma L
φ^0_λ(M'')φ^0_μ(M'')|_H ≠ 0, and so its deformation φ_$\lambda\varphi$_μ(M'')|_H is ≠ 0 at generic q. ■

So, for every (λ, μ), *some* structure constant c^•_{$\lambda\mu$} acquires a long-root pole as soon as N ≥
↪ max(|λ|,|μ|) + 2,
which is exactly when a jump-≥2 ν exists. The per-ν converse is the hard part.

5. The first-order residue formula -- PROVED

Theorem 7. For every N, every triple, and every a<b:
Res_{H_ab} c_1^ν_{$\lambda\mu$} = γ · Res_{H_ab} ℓ^0_ν(1_{⊇ab} · P), where P := φ^0_$\lambda\varphi^0$_μ.
Here Res f := (h f)|_H, ℓ^0_ν is the ν-coordinate in the basis φ^0, and 1_{⊇ab} is the indicator of {M ⊇
↪ {a,b}}.

Proof. By (1.1), c_1 = $\ell^0(\Phi)$. Φ ∈ O_H^{2^N}: δ is Laurent in w, and c_0 is regular on H by Prop 5 at q=0,
↪ because
P ∈ L_1 (Corollary 4(i)). By Prop 5 the residue depends only on the defects of Φ. By Corollary 4:
Φ(M) − Φ(M'') = γP(M'') + γP(M'') − $\gamma\Sigma$_κ c_0^κ φ^0_κ(M'') = γP(M'') on H. The function 1_{⊇ab}P has defect
P(M) − 0 = P(M'') on H at every M ⊇ ab. ■

Checked against direct c_1 on all 167 (triple, pair ⊆ ν) at N=3 (`testresq0.py`).
Note: ℓ^0_ν only sees values on ↓ν, because the inverse of a matrix triangular for the order is again
↪ triangular.

6. The top-two theorem -- PROVED

Prop 8 (forced heads). Let a<b, let κ = κ''∪{a,b} and M = M''∪{a,b} with κ'', M'' ⊆ [a−1]. Then
φ^0_κ(M) = e(M'') φ^0_{κ''}(M''), where φ^0_{κ''}(M'') is formula (2.1) for subsets of [a−1], and
e(M'') = (1−w_a^{-2})(1−w_b^{-2})(1−1/(w_aw_b)) Π_{c∈M''}(1−1/(w_cw_a))(1−1/(w_cw_b))
· Π_{c∈[a−1]\M''}(1−w_c/w_a)(1−w_c/w_b) · Π_{a<c<b}(1−w_c/w_b).

Proof. In (2.1), label b needs a head w_m with m ≥ b, otherwise the term vanishes identically. The only such
element of M is b. Label a then needs m ≥ a with m ≠ b, which forces m = a. In the cross factors, every c ∈
↪ M''
(head or tail) meets the heads w_a, w_b through the same label-free expression
(1−1/(w_cw_a))(1−1/(w_cw_b))/((1−w_c/w_a)(1−w_c/w_b)). Its denominators cancel the factors k = c of
Π_{k<a}, Π_{k<b}, and (1 − w_a/w_b) cancels the pair denominator. What remains is exactly the [a−1]-formula
for κ'' on M''. ■

On H: e = h·e', and the identity (1−1/(w_cw_a))(1−w_a/w_c) = w_c^{-2}(1−w_c/w_a)(1−w_aw_c) gives
(6.1) e'(M'')|_H = −γ·K·Π_{c∈M''} w_c^{−2}, K := Π_{c<a}(1−w_c/w_a)(1−w_aw_c) · Π_{a<c<b}(1−w_aw_c) ≠
↪ 0.

Theorem 9 (top-two). Let a<b be the two largest elements of ν, put ν'' := ν\{a,b}, and suppose ν'' ≥ λ
↪ and
ν'' ≥ μ. Then
Res_{H_ab} c_1^ν_{$\lambda\mu$} = −K^{−1} · Ξ, where Ξ := [φ^0_{ν''}](M'' ↦ L(M'') φ^0_λ(M'')φ^0_μ(M'')) ≠ 0,
with L(M'') := Π_{c∈M''} w_c^2, and the coordinate taken in the q=0 system on subsets of [a−1]. In particular
c_1^ν has a simple pole on H_ab, so **c^ν_{$\lambda\mu$} is not Laurent** (§2).

Proof.
1. U := {M $\leq \nu$: a, b $\in$ M} = {M''$\cup$ab : M'' $\subseteq$ [a$-$1], M'' $\leq \nu$''}, and U is an up-set of $\downarrow\nu$. Reason: M $\leq \nu$ forces
the top two elements of M to be $\leq$ (a, b), and a, b $\in$ M forces them to equal (a, b). For M $\leq \tilde{M} \leq \nu$ the top two
of $\tilde{M}$ are squeezed between those of M and those of ν.
2. On $\downarrow\nu$ we have 1_{$\supseteq$ab} = 1_U. Functions on $\downarrow\nu$ supported in U are spanned by {φ^0_κ : $\kappa \in$ U}, which is triangular on U. So, by Theorem 7, Res c$_1$^ν = γ Res ℓ^U_ν(P|_U), with ℓ^U the coordinates in {φ^0_κ|_U}.
3. By Prop 8, ℓ^U_ν(F) = ℓ^{[a$-$1]}_{ν''}(F($\cdot\cup$ab)/e). The [a$-$1]-basis involves only w_1, ..., w_{a$-$1}, and its diagonal factors are nonzero, so it is regular and invertible on H. Hence
Res ℓ^U_ν(P|_U) = ℓ^{[a$-$1]}_{ν''}(P($\cdot\cup$ab)/e')|_H. Since P(M''$\cup$ab)|_H = P(M'')|_H (Corollary 4), (6.1) gives Res c$_1$^ν = $\gamma\cdot(-1/(\gamma$K$))\cdot\Xi$.
4. $\Xi \neq 0$. Take w_c = t^{$-$e_c} with e_1 < ... < e_{a$-$1}. By Lemma L the [a$-$1]-matrix tends to the zeta matrix, so its
inverse stays bounded and [ν'', ν'']-entry $\to$ 1. Also P(M'') $\to$ [M'' $\geq \lambda$][M'' $\geq \mu$], and for M'' < ν'' we have
L(M'')/L(ν'') $\to$ 0, because Σ_{c$\in$M''} e_c < Σ_{c$\in\nu$''} e_c under top-aligned domination. So Ξ/L(ν'') $\to$ P-limit
at ν'' = 1. ■

Checked: the formula $-$K$^{-1}\Xi$ matches direct c$_1$ on all 105 triples at N = 2, 3 (`toptwo.py`), zero cases included.

Coverage: every ($\varnothing$, $\varnothing$, ν) with |ν| $\geq$ 2; the N=2 witness (residue -1, computed by hand); 6 of 13 jump-$\geq$2 triples
at N=3. The other seven (λ or $\mu \in$ {{2},{3}}, ν = {1,2,3}) have residue 0 on the top pair (2,3), as the formula predicts. They fall to the next theorem, at pair (1,2).

Theorem 9' (single defect). Suppose some a<b in ν has ν'' := $\nu\backslash${a,b} $\geq \lambda$, μ, and no M < ν has a, b $\in$ M and
M'' $\geq \lambda$, μ. Then Res_{H_ab} c$_1$^ν = $\gamma\cdot\rho$_{$\nu\nu$}$\cdot$P(ν'')|_H $\neq$ 0, where ρ_{$\nu\nu$} = Res_H(1/φ^0_ν(ν)).
Proof. By Prop 5 and Theorem 7, Res = γ Σ_{M$\supseteq$ab} ρ_{νM}P(M'')|_H with ρ_{νM} = Res ℓ^0_ν(e_M). That vanishes unless
M $\leq \nu$, and P(M'') = 0 unless M'' $\geq \lambda$, μ (M3). Only M = ν survives. ρ_{$\nu\nu$} $\neq$ 0 because φ^0_ν(ν) has a simple zero on
H (M4), and P(ν'')|_H $\neq$ 0 by Lemma L, since ν'' $\not\ni$ a, b. ■

Coverage of Theorems 9 $\cup$ 9' (`coverage.py`, purely combinatorial): N=2: 1/1; **N=3: 13/13**; N=4: 111/127; N=5: 792/1136; N=6: 5117/9748. The 16 uncovered N=4 triples are (λ,μ) $\subseteq$ {$\varnothing$,{1},{2},{3},{4}}, $\nu \in$ {134, 124, 1234}.
15 of them have a c$_1$ pole numerically; the 16th is the counterexample of §7.

Full-q version. The forced-head argument uses only the factor Π_{k<i}(u $-$ w_k), which carries no q. Together
with QC this gives Res_{H_ab} c^ν = (g^2 $-$ g)$\cdot$[ψ_{ν''}](φ_$\lambda\varphi$_μ(M'')/e'^{(q)}(M''))|_H at full q, where ψ is the
reduced system. This is sketched, not written out. Its q $\to$ 0 limit is Theorem 9.

6bis. (d) and Laurentness of c$_0$ -- PROVED, all N

Lemma C (collapse; Clio's Lemma 3, re-derived). For all q, with τ as in Clio:
(i) on {w_a = 1}, for a $\in$ M: $\tilde{A}$_{M,I} = $\tilde{A}$_{M\a,I};
(ii) on {w_a = w_b}, a<b, for b $\in$ M, a $\notin$ M: $\tilde{A}$_{M,I} = $\tilde{A}$_{(M\b)$\cup$a,I};
(iii) $\tilde{A}(-w) = \tilde{A}(w)$, and hence (i) also holds on {w_a = $-$1}.
Proof. (i), (ii): pad to a common n. The two specialisation tuples then have the same multiset, and F and E^C are
symmetric in u (Motegi L455-458, L1057). (iii): under (u,w) $\mapsto$ ($-$u,$-$w), the factors of F change sign with total
exponent r(N$-$1) + (n$-$r)N + (n$-$r) + n(N+n) $\equiv$ n(n+1) $\equiv$ 0 (mod 2). The heads give (N$-$i)+(i$-$1), the tails N, (βu$-\gamma$) = u
gives 1, and the pair factors are invariant. Every factor of E^C is invariant. Then apply (i) to $-$w. ■

Lemma C' (collision regularity at q=0). Let D be one of {w_a = $\pm$1}, {w_a = w_b}, with collapse map τ on $\mathcal{D}$.

If f is a vector of Laurent polynomials with f(M)|_D = f(τM)|_D for all M $\in \mathcal{D}$, then ℓ^0(f) is regular on D.
Proof. This is Clio's thm:collision induction, verbatim, with f in place of φ_$\lambda\varphi$_μ. It needs: τM < M (Clio);
φ^0 has the collapse property (Lemma C at q=0); and A_0_{ν,ν} vanishes on D to order exactly 1 for $\nu \in \mathcal{D}$ and to order 0 otherwise. The last holds because the S-factor (1 − w_a^{-2}) is present iff a $\in \nu$, and the A-factor (1 − w_a/w_b) is present iff a $\notin \nu \ni$ b, each once. ■

Theorem 10. For all N: (i) c_0^ν_{$\lambda\mu$} is a Laurent polynomial in w for every triple; (ii) if jump $\leq$ 1, then
c_1^ν_{$\lambda\mu$} is a Laurent polynomial in w. (iii) For every triple, every non-monomial pole of c_1 lies on a long-root
divisor, and it is simple.

Proof. ℓ^0 = A_0^{-1} = adj/det, and det A_0 = Π_I A_0_{I,I} is a product of L, S, A factors and monomials. So for a
Laurent f, the only possible non-monomial poles of ℓ^0(f) lie on L, S, A divisors. The vectors P = φ^0_$\lambda\varphi^0$_μ and Φ
(1.1) are Laurent. (A_1 = ∂_q$\tilde{A}|_0$ is Laurent, because $\tilde{A} \in$ C[q][w^±] localised at units 1 − q·monomial. And c_0 is
Laurent by (i), which is proved first.)
- S, A divisors: P and Φ have the collapse property. φ^0 and δ have it by Lemma C (differentiate in q), and c_0 is
  regular on D. Lemma C' applies.
- Long roots: c_0 is regular, because P $\in L_1$ (Corollary 4, Prop 5). For c_1, Theorem 7 gives Res = γ Res ℓ_ν(1_{$\supseteq$ab}P).
  The residue is Σ_{M$\supseteq$ab, M$\leq\nu$} ρ_{νM}·P(M'')|_H. When jump $\leq$ 1, every such M has |M''| $\leq$ |ν| − 2 < max(|λ|,|μ|),
  so P(M'') = 0 by (M3). Simplicity of the poles comes from Prop 5. ■

So (d) is the first-order shadow of Clio's thm:main, and **at first order it needs no Gap q**: the (Q)-family is
not a divisor at q = 0. Theorem 10(i) also turns Rick's Day-242 census "195/195 Laurent at q = 0" into a theorem.

7. Why (c) fails: the high-block cancellation (N = 4)

Triple (λ, μ, ν) = ($\varnothing$, {3}, {1,2,4}): jump 2, and Lemma-comb pair (1,2).
- Numeric engine (`num4.py`, independent of the §3-6 theory, Motegi summand with q symbolic): c_1 has no pole on any
  of the six H_ab. Confirmed by the Theorem 7 route (`resnum.py`) and on two further seeds.
- Full q (`fullc.py`): c^{124}_{$\varnothing$,3} has a simple pole on H_{12} only. Its residue is
  $7q^2(q-1)^6$·(quintic in q)/const: **q-valuation 2**. So Clio's converse (and her Conjecture "exact pole divisor")
  survives for this triple, but the pole is invisible at q^1.
- Mechanism. The defect set is D = {123, 124}, an interval and an up-set of $\downarrow\nu$. The pair (1,2) has an empty low
  block and a high block {3} $\leftrightarrow$ {4}. The ν-coordinate is a divided difference over the high block, and at q=0 the
  restricted function $\varphi^0{}_3$|_D is proportional (through the label-free factor Γ) to the basis element φ^0_{123}|_D.
  So the q^1 residue is 0. In Theorem 9 it is the *low* block's weight L(M'') that forces non-vanishing. A
  high block alone gives a divided difference that can vanish at q = 0.

Also: (b) fails because φ^0 is (by every check: diagonal = the codim-many tangent weights, support = containment of
strict partitions, GKM) the K_T(LG(N,2N)) Schubert basis. In cohomology $\sigma_{21}\cdot\sigma_{43} = \sigma_{4321}$ in LG(4,8) is a jump-2
product, and c_0^{1234}_{12,34} = $1/(w_3^2 w_4^2)$ is its equivariant K-lift (`c0_C4.txt`). The identification with K_T(LG)
is a **hunch** (strong): nothing above uses it.

8. Computational record

- q=0, N=3: no jump-$\geq$2 $c_0 \neq 0$; all c_0 Laurent. N=4: 17 jump-$\geq$2 $c_0 \neq 0$ (list in `c0_C4.txt`), all c_0 Laurent.

- c_1, N=2,3 (symbolic, `c1N.py` with engine3, validated vs engine2 vs the Day-242 engine): every jump-$\geq$2 triple has a
  c_1 pole (H_{12} always among them); every jump-$\leq$1 c_1 is Laurent.
- N=4 numeric (`num4.py`), H_{12}: jump 2: 107 poles, 1 none; jump 3: 18/18; jump 4: 1/1; jump $\leq$ 1: 0 poles in 599.
  Remaining pairs: `num_N4_{13,14,23,24,34}.txt`, see the addendum below.
- Controls: QC fails off H. The ratio is I-independent only on H (a single-I comparison would have been blind, which is why every I was compared). The top-two formula predicts zeros on the non-covered triples, and they are
  zero.

9. Gaps (precise)

1. Per-ν converse for triples not covered by Theorem 9. Needed: for some pair (a,b), the full-q residue $(g^2 - g)\cdot\mathrm{Res}\ \ell_\nu((1-1/g)\dots\ ) \neq 0$. That is a statement about the *q-valuation* of a high-block divided difference, and order q^1 is provably insufficient. Candidate tools: (i) the q $\leftrightarrow$ 1/q symmetry of g, which suggests the q$\to\infty$ expansion, where $\tilde{A}\cdot$q^{$-$(dim $-$ codim I)} has a finite limit (probed at N=2), may cover high
   blocks; (ii) the block factorisation A|_V = diag(Γ)(A^{high} $\otimes$ A^{low}), valid when the pair is consecutive in
   ν and V is an interval with fixed block sizes (sketched, verified on the N=6 example (5,5,{1,3,4,6}), where the
   poles are exactly Clio's predicted pairs).
2. (d) is closed (Thm 10). Not closed: (d) at full q, which is Clio's thm:main and still needs her Gap q.
3. Theorem 6 and Prop 5 need "no factor of $\tilde{A}$_{I,I} other than the long root vanishes on H_ab", which I read off
   (M4) and Clio's eq:diag (and checked at N $\leq$ 3). Fine, but noted.

Addendum (same session): full-q converse data and the q = 0 / q = ∞ dichotomy

- **N=4, full q, H_{12} (`dich.py`, `dich_N4_12.txt`): all 127 jump-$\geq$2 triples have a pole on H_{12}.** So Clio's converse
  holds at N=4 for every triple (computed; her census stopped at N=3).
  - q-valuation of the residue at q=0 is 1 for 126 triples and 2 for ($\varnothing$,{3},{1,2,4}).
  - **Degree at q=∞ equals e$-$1 for 127/127**, where e = dim + codim ν $-$ codim λ $-$ codim μ (dim = N(N+1)/2, codim = sum
    of elements). Since $\hat{\varphi}$_I := q^{$-$(dim$-$codim I)}φ_I has a finite limit φ^∞ (exponent derived from Motegi's formula and
    checked at N=2), this says the residue is *first order at q=∞* in every case.
  - N=3: 13/13 first order at both ends.
- **N=5, q^1 only (`n5scan.py`):** (c) fails for exactly 12 of 1136 jump-$\geq$2 triples, all with "high-block" data:
  $\lambda=\varnothing$ (and one (λ,μ) = ({1},{4})), with μ supported on the top of ν. The other 332 triples not covered by Thms 9/9'
  have a q^1 pole numerically.
- **N=5, the q^1 failures at full q (`qinftest.py`, so far 5/12, rest running):** every predicted pair has q-valuation 2
  at q=0 and is first order at q=∞.

Hunch (registered): every jump-$\geq$2 residue on H_{12} is first order at q=∞. True on 127/127 at N=4. If it holds,
the converse would follow from a q=∞ mirror of Theorem 7 plus a non-vanishing argument at q=∞. Note that at q=∞ the
QC twist, the triangularity support and the forced heads are unchanged; what changes is the pair/tail ordering in the
leading coefficient, which is reversed. Unlike (2.1), each term also carries an assignment-dependent monomial Π_j h_j^{κ_j+n$-$2N$-$1} $\Pi_{k>\kappa_j}$ w_k $\cdot$ Π_t t^{r$-$2N}. So a q=∞ "Lemma L" needs a genuine dominant-term analysis. That is
the next PROVE on this arc.
- **φ^∞ in the generic regime** (N=3, `phiinf.py`, w_i = t^{$-$e_i} with e = (1,3,9)): every entry with M $\geq$ κ is
  $\pm$(monomial) $\neq$ 0, with sign exactly (-1)^{codim κ}. Entries with M $\not\geq$ κ are 0. But the exponents are **not** of
  row$\times$column form (row {1,3} minus row {3} = 6, 6, 4, 4), so φ^∞ is not "monomial$\cdot$zeta". The q=∞ analogue of

```
Lemma L's non-vanishing survives, but Theorem 9's dominant-term step needs a tropical (min-plus) inverse at
↪  q=∞.
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